

Sulphur content of marine fuels Inspection and enforcement

22-05-2023

Workshop SECA enforcement



Purpose of this presentation

- Short look at the regulations!
- How does DMA conduct sulphur inspections?
- On which vessels do we inspect!
- What are the result and consequences for violations? (MoEoD)



Presentation

Wytse van der Molen

Senior Ship Surveyor

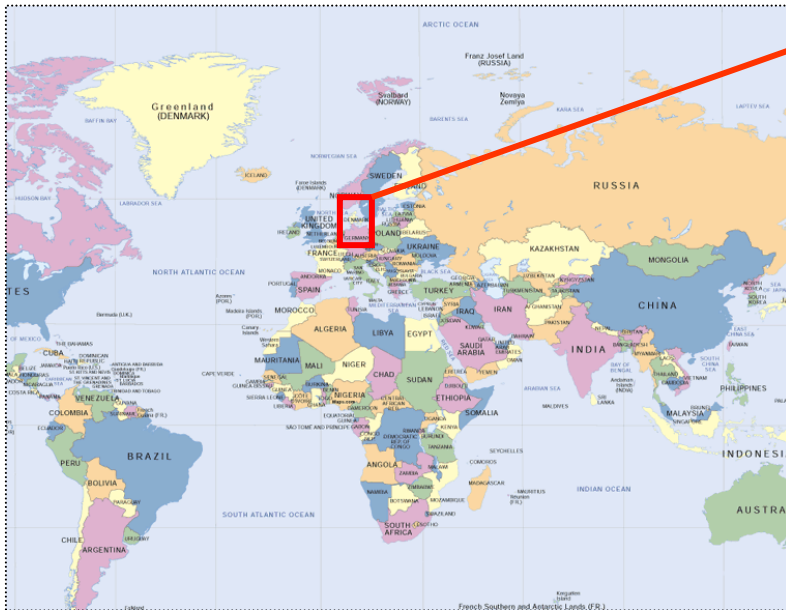
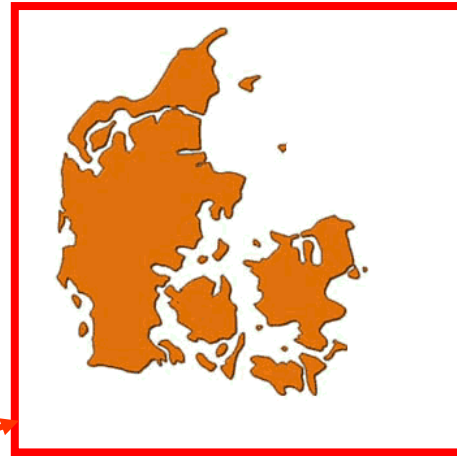
DMA 2008 - ?

Flag State & Port State Control Officer

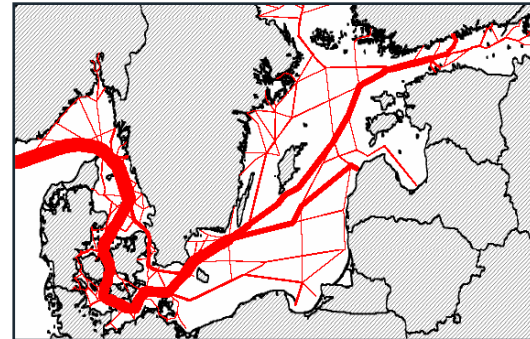


DANMARK

400 KM

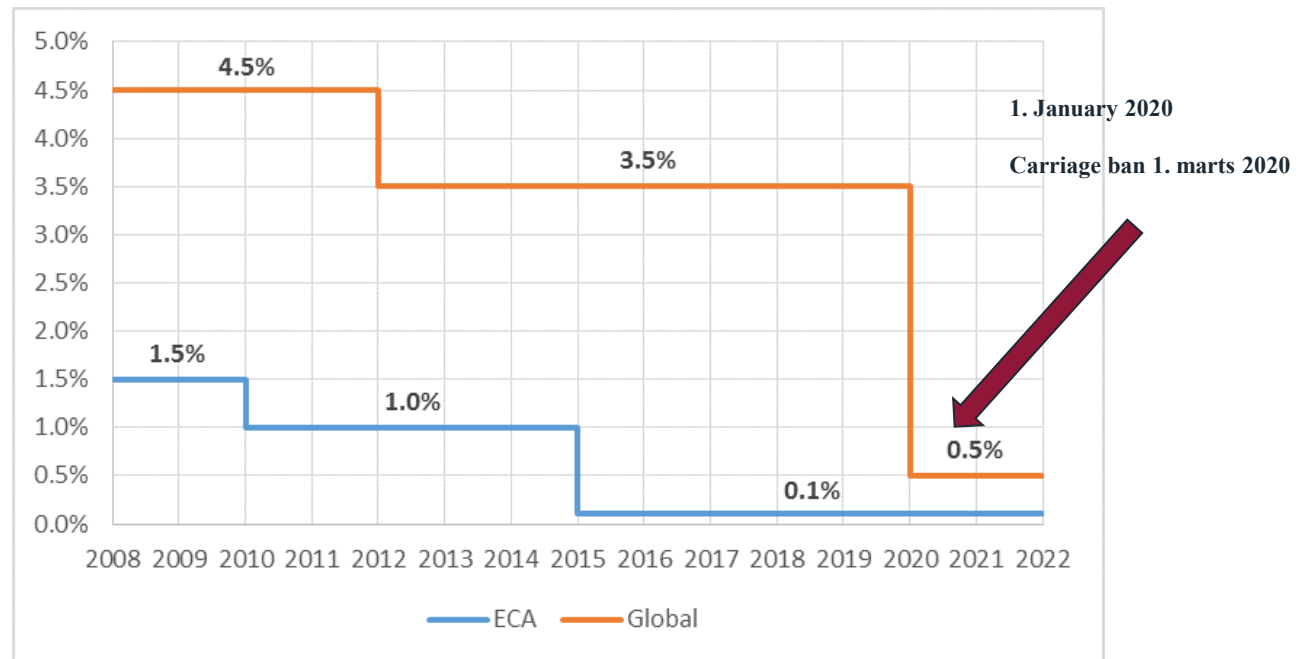


- > 1400 islands
- 8750 km of coastline



Regulation to prevent Sulphur oxides emissions

- MARPOL – Annex VI **Prevention of air pollution from ships** – Ch. III, reg. 14
Sulphur Oxides (SOx) and particulate matter
- Applies to all ships – **regardless of size** – domestic and international



Sulphur Emmission Control Area



The Danish system

Regulation responsibility

EU Sulphur Directive,
Svovl Bek.



Fine or prosecution

Inspection is delegated



SOLAS
MARPOL



Inspection of ships



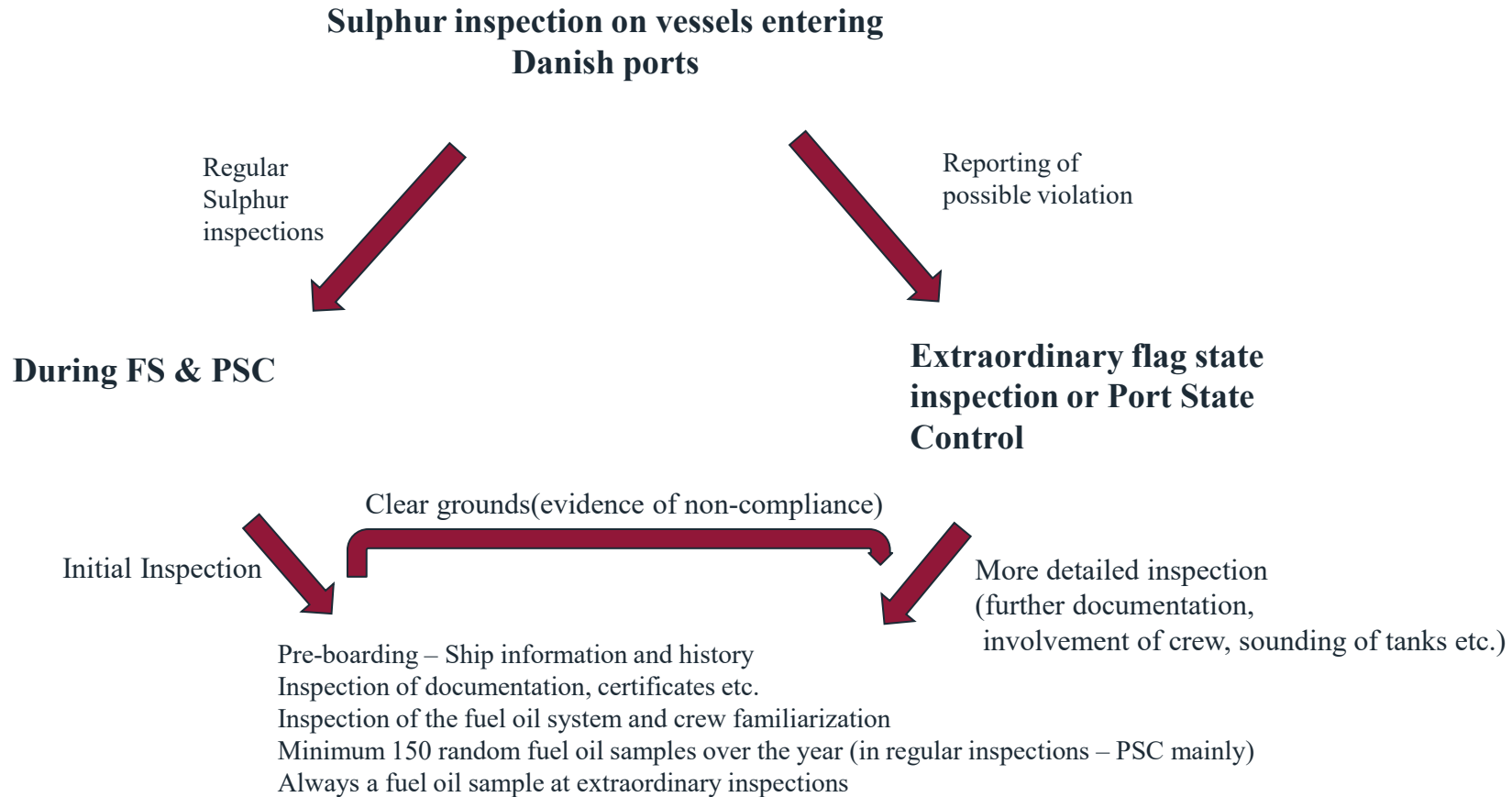
Compliance failure

Reporting

Based on sample or
observations
EU central database



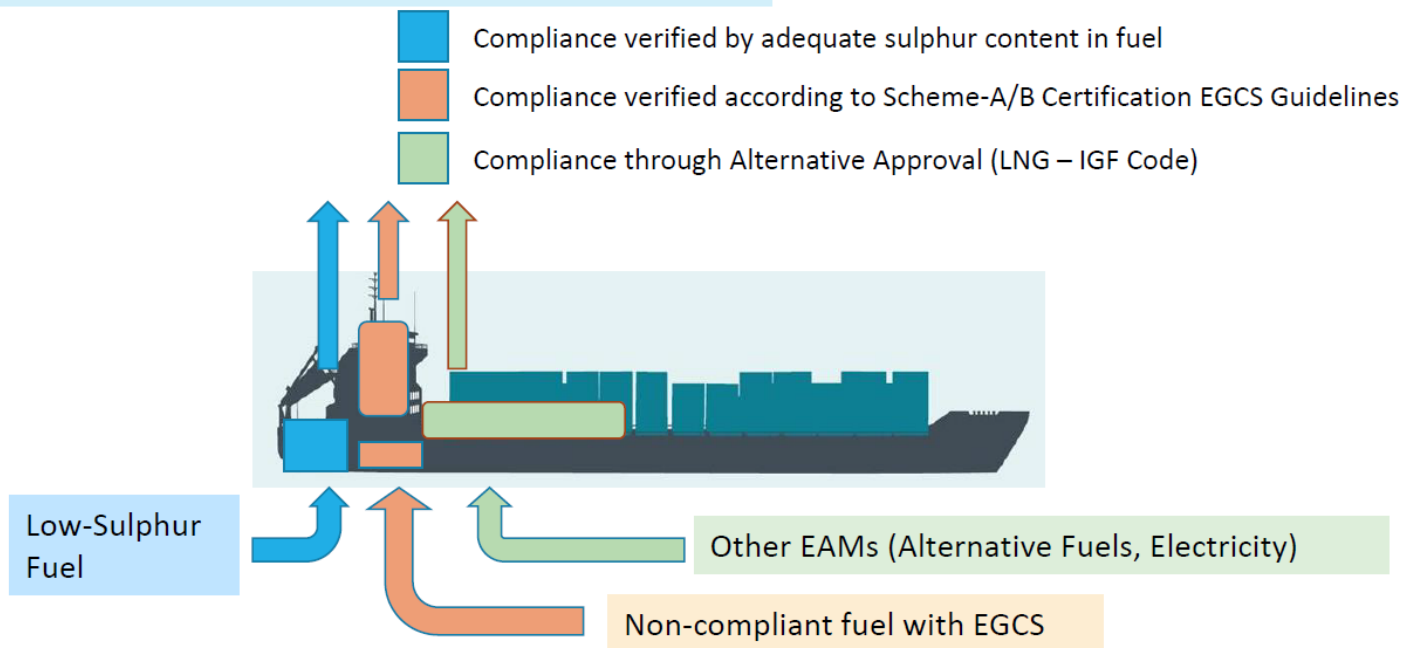
The enforcement



How to comply

Emission Abatement Methods

1. Primary method - Low-Sulphur Fuel
2. Alternative Arrangements



Sulphur inspection on board

Vessel using Low Sulphur Fuel

- Check IAPP certificate (Scrubber Y/N?)
- Correct BDN > MARPOL/IV/18
- Sealed sample
- Is there fuel > 0,1% on board
- If yes: correct documentation in Oil Record Book
- Procedures for change over to low sulphur before entering SECA



Fuel Samples

Minimum 150



Selection of ships:

1. Missing BDN or sample
2. Incorrect entries ORB
3. Inconistencies when using alternative arrangements
4. Ship arriving from outside SECA

Sample collection

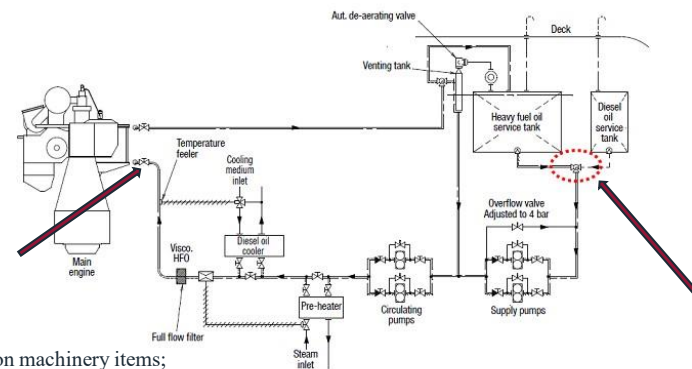
- 2016 Guidelines for on-board sampling for the verification of the Sulphur content of the fuel oil used on board ships' (IMO MEPC.1/Circ.864 Rev. 1)



In Denmark/EU - 3 single spot samples

- 1 for the ship
- 2 for analysis

- be easily and safely accessible,
- take into account different fuel grades being used for the fuel-oil combustion machinery items;
- be downstream of the fuel in use from the service tank,
- be as close to the fuel inlet of the fuel-oil combustion item as feasible and safely possible taking into account the type of fuels, flow-rate, temperature, and pressure behind the selected sampling point,



Important to make sure that there no "in" connections between the service/day tank and the engine

Valves in right position (not half closed/open)

Signs of dismantling or tampering

"Magical" pipes

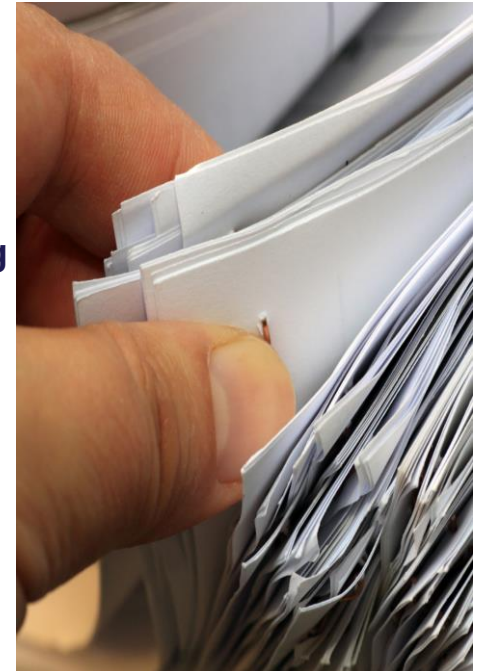
Crew familiar with system and can demonstrate the change over procedure – if relevant.



Send to laboratory for analysis – ISO 8754

Inspection – PSSC 93 – Guidance for PSCOs

1. Fuel oil with correct sulphur content
2. Since 1 March 2020, the sulphur content of fuel oil used or carried for use on board a ship shall not exceed 0.50% m/m unless approved alternative methods are installed
3. Appropriate approval for any installed exhaust gas cleaning systems, or equivalent means, if required
4. There are bunker delivery notes (BDNs) and associated samples
5. Written procedures fuel oil change over operations
6. Familiar with essential fuel oil management procedures



Alternativ arrangements - EGCS

WHAT DO WE LOOK AT:

- Appropriate approval for any installed exhaust gas cleaning systems, or equivalent means
- EGCS monitoring as required, records maintained
- Correct BDN < 3.5% & oil samples in board

2.3 Sulphur oxides (SO_x) and particulate matter (regulation 14)

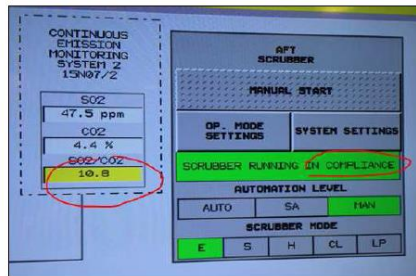
2.3.1 When the ship operates outside of an Emission Control Area specified in regulation

14.3, the ship uses:

- .1 fuel oil with a sulphur content as documented by bunker delivery notes that does not exceed the limit value of:
3.50% m/m (not applicable on or after 1 January 2020); or ☐
0.50% m/m, and/or ☐
- .2 an equivalent arrangement approved in accordance with regulation 4.1. as listed in 2.6 that is at least as effective in terms of SO_x emission reductions as compared to using a fuel oil with a sulphur content limit value of:

Inspection of EGCS

- The inspector must rely on certification, documentation, the continuous monitoring and his/hers professional judgement



Fuel oil sulphur content (% m/m)	Ratio emission SO ₂ (ppm)/CO ₂ (% v/v)
4.50	195.0
3.50	151.7
1.50	65.0
1.00	43.3
0.50	21.7
0.10	4.3

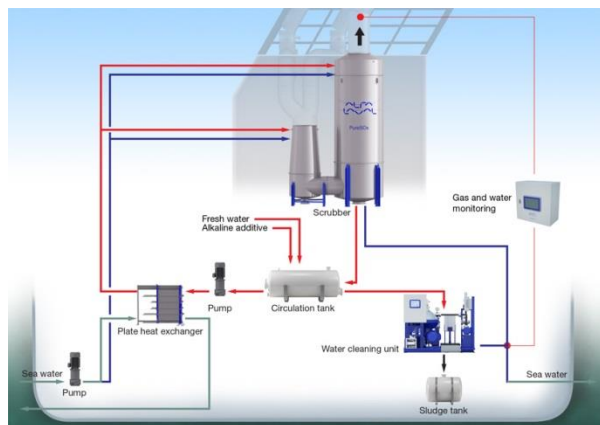
Note: The use of the ratio emissions limits is only applicable when using petroleum based distillate or residual fuel oils. See appendix II for application of the ratio method.

- EGCS monitoring records have been retained
- System is properly functioning and in operation
- On-board Monitoring Manual (OMM) available
- Master or crew are familiar with the correct operation of an EGCS
- There is more please note **RESOLUTION MEPC.321(74)**

Alternative arrangement - EGCS

The PSCO should check:

- If the ship's Administration has allowed an alternative arrangement that may be equivalent to the standards in MARPOL Annex VI, regulations 13 and 14 (MARPOL Annex VI, regulation 4.1).
- If such an alternative arrangement has been communicated to the Organization/IMO (MARPOL Annex VI, regulation 4.2).



**SUPPLEMENT TO
INTERNATIONAL AIR POLLUTION PREVENTION CERTIFICATE
(IAPP Certificate)**
RECORD OF CONSTRUCTION AND EQUIPMENT

In respect of the provisions of Annex VI of the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (hereinafter referred to as "the Convention").

Notes:

1. This Record shall be permanently attached to the IAPP Certificate. The IAPP Certificate shall be available on board the ship at all times.
2. The Record shall be at least in English, French or Spanish. If an official language of the issuing country is also used, this shall prevail in case of a dispute or discrepancy.
3. Entries in boxes shall be made by inserting either a cross (x) for the answers "yes" and "applicable" or a (-) for the answers "no" and "not applicable" as appropriate.
4. Unless otherwise stated, regulations mentioned in this Record refer to regulations of Annex VI of the Convention and resolutions or circulars refer to those adopted by the International Maritime Organization

1 Particulars of ship

- 1.1 Name of ship
- 1.2 IMO number
- 1.3 Date on which keel was laid or ship was at a similar stage of construction
- 1.4 Length (L) in meters

Completed only in respect of ships constructed on or after 1 January 2016, which are normally

2.6 Equivalents (regulation 4)

The ship has been allowed to use the following fitting, material, appliance or apparatus to be fitted in the ship or other procedures, alternative fuel oils, or compliance methods used as an alternative to that required by this annex:

System or equipment	Equivalent used	Approval reference
4 X Main Engine Scrubber system	EGCS STI2015 In-line open loop scrubber	CONV5023

International Maritime Organization

17 December 2014

Our reference:

Case: [REDACTED]

Maritim Regulering og Besættning

[REDACTED] (SFS)

Acceptance of Equivalent Compliance Method under Regulation 4.1 of Annex VI of the MARPOL Convention

This is to certify that the Danish Maritime Authority on the 25 May 2012 has accepted the Alfa Laval Puresox exhaust cleaning system, type hybrid multiple inlet, as an equivalent compliance method to that specified in regulation 14 of MARPOL Annex VI to be utilized onboard [REDACTED] in accordance with Regulation 4.1 MARPOL Annex VI.

The system has been certified to comply with MEPC 170(57) Scheme B.

DANISH MARITIME AUTHORITY

Carl Jacobsens Vej 31
DK-2500 Valby
Denmark

Tel. +45 72196000

Fax +45 72196001

C/R.no. 29 83 16 10

EAN-no. 5798000023000

dma@dma.dk

www.dma.dk

MINISTRY OF BUSINESS AND GROWTH

Yours sincerely

[REDACTED]

[REDACTED] (SFS)
Special Adviser

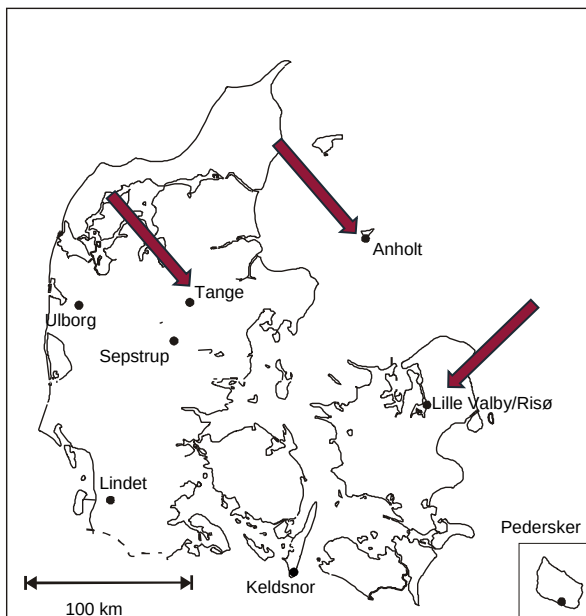
Carriage Ban

- DMA checks last 3 BDN on board!



Effect of the Sulphur Regulation on Air Quality in Denmark

- Measuring stations



	Anholt	Risø	Tange
	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$	$\mu\text{g}/\text{m}^3$
Average 2011-2014	0,33	0,34	0,22
2015-2016*	0,13	0,17	0,10
Reduction	62 %	50 %	56 %

*) Data for 2016 are preliminary

Significant reduction

